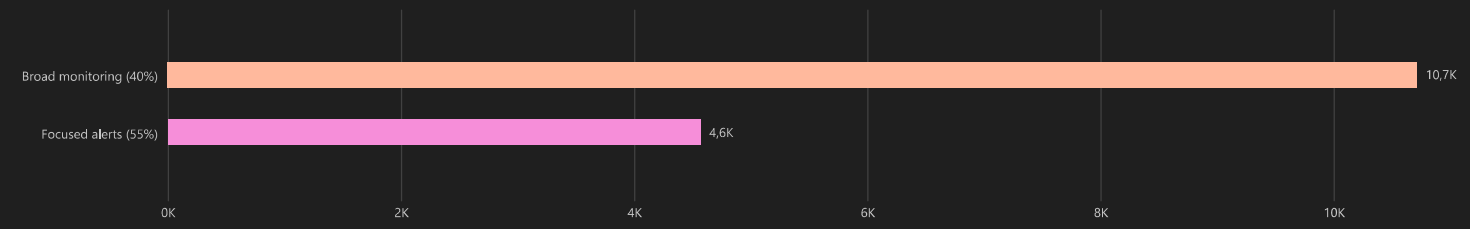
SUPPLY CHAIN RISK | EXECUTIVE OVERVIEW

DataCo dataset · Test period: 1 Jul 2017–31 Jan 2018

Test Orders	Positive Label Orders	Focused Precision	Broad Recall
13,07K	8K	88,84 %	87,71 %

Orders Flagged by Risk Policy

Sum of orders_flagged by Policy Display



Monthly Observed Positive-Label Rate

by Year, Quarter, Month and Day

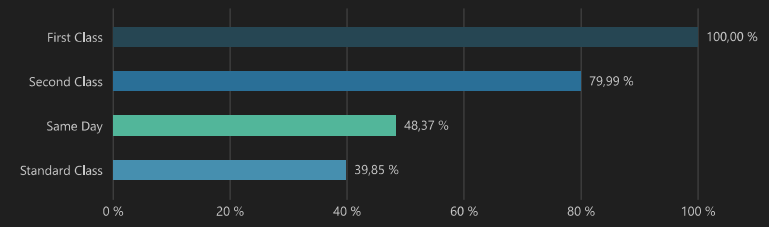


DELIVERY RISK DRIVERS

Historical patterns across shipping modes, markets, regions and time.

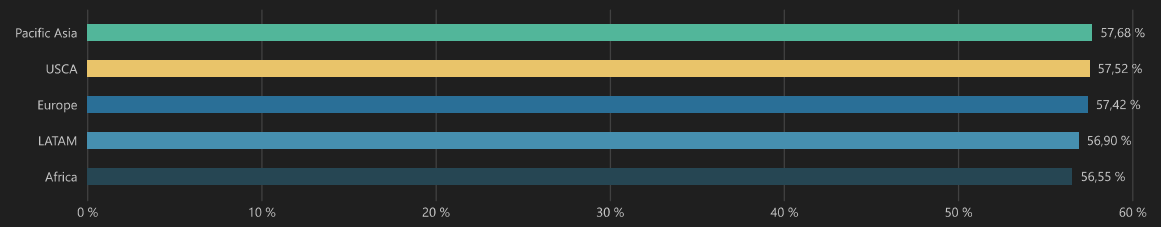
Historical Positive Label Rate

by shipping_mode



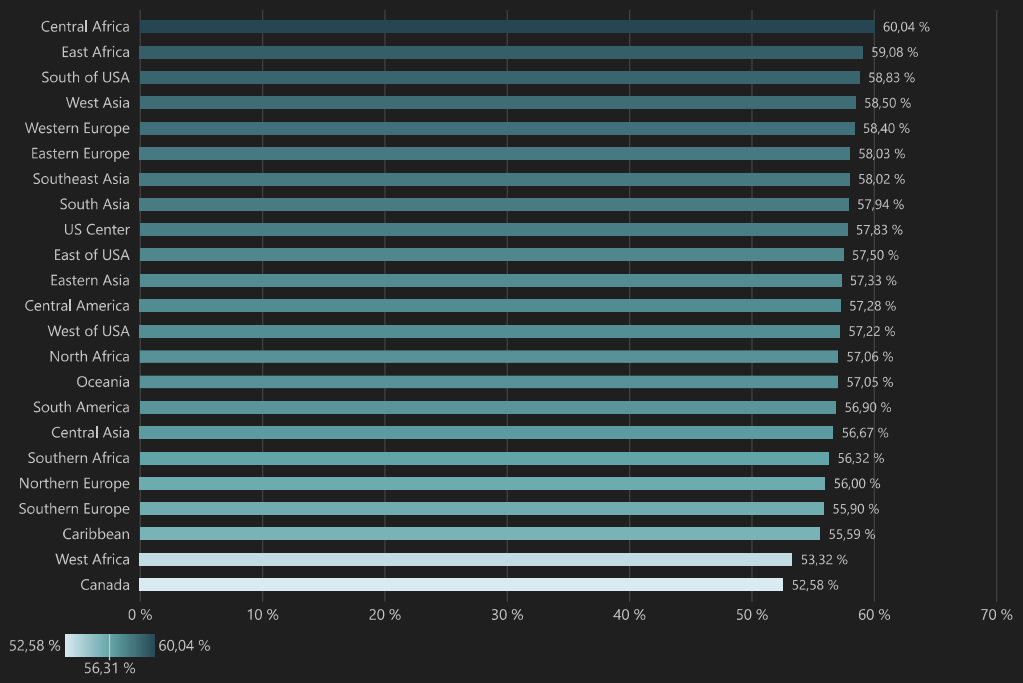
Recorded Positive-Label Rate by Market

Historical Positive Label Rate and Historical Orders by market



Recorded Positive-Label Rate by Destination Region

Historical Positive Label Rate and Historical Orders by destination_region

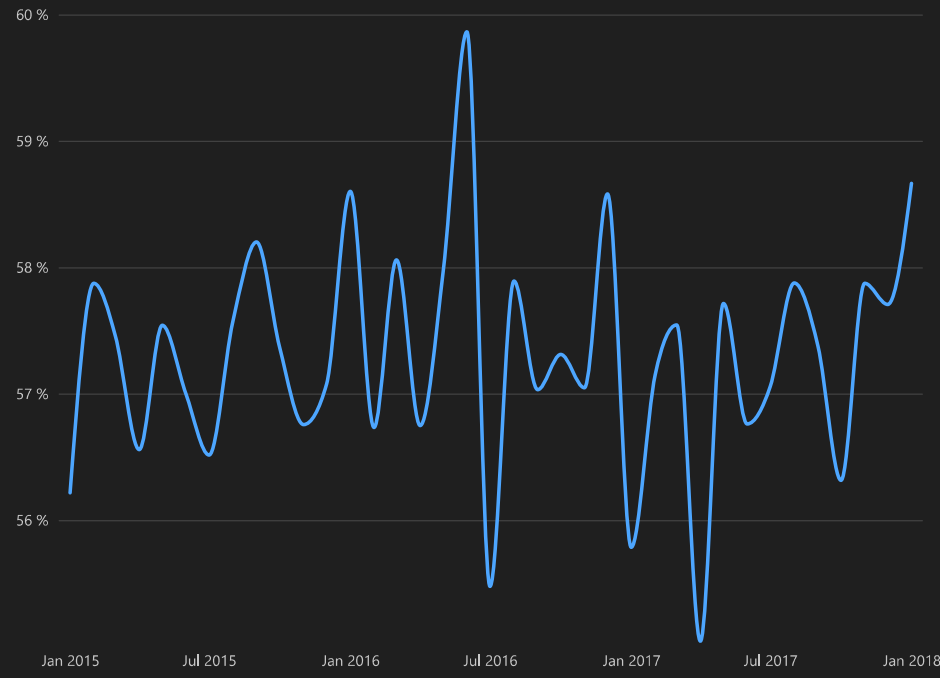


Key insight

Recorded positive-label rates vary across destination regions, but the differences should be interpreted alongside order volumes. Regions with fewer orders may have less stable rates. These descriptive patterns do not establish the causes of delivery delays.

Monthly Recorded Positive-Label Rate

Historical Positive Label Rate and Historical Orders by Order Month Start



The monthly recorded positive-label rate provides historical context for delivery-risk patterns from January 2015 through January 2018. These trends describe recorded outcomes in the dataset and should not be interpreted as evidence that time itself caused changes in delivery risk.

FINANCIAL SCENARIO ANALYSIS

Comparing monitoring policies under hypothetical delay-cost and mitigation

Broad Hypothetical Cost Avoided
31,83K

Focused Hypothetical Cost Avoid...
19,73K

Broad Missed-Cost Proxy
19,89K

Focused Missed-Cost Proxy
65,38K

Assumed Mitigation Effectiveness (%)

- (●) Deselect all
- (●) 10
- ( ) 25
- (●) 40

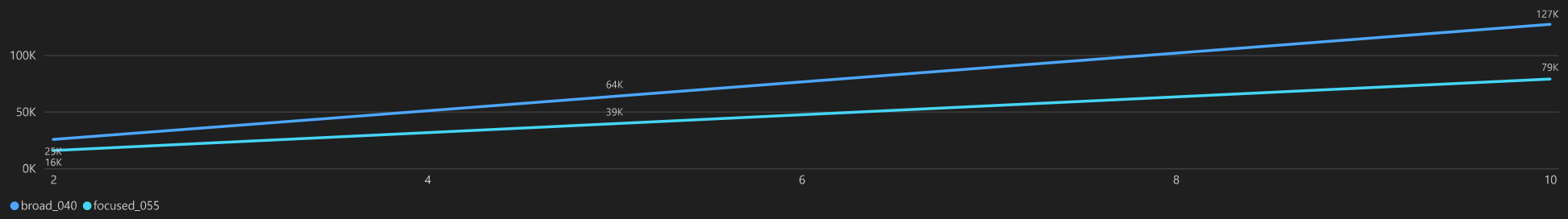
Monitoring Policy — Financial Comparison

Policy Display	orders_flagged	flagged_order_value	order_value_from_false_alerts	hypothetical_cost_avoided	missed_late_order_cost_proxy
Broad monitoring (40%)	10714	3997721,58	1505835,05	31834,06	19.888,29
Focused alerts (55%)	4570	1777944,78	195932,24	19734,12	65.381,98

Broad monitoring flags more orders and captures more potential exposure, but creates substantially more alert workload. Focused monitoring flags fewer orders and has lower false-alert exposure, but leaves more positive-label order value unflagged. The preferred policy depends on operational capacity and the costs of missed delays

Hypothetical Cost Avoided Under Different Scenarios

Sum of hypothetical_cost_avoided by hypothetical_delay_cost_rate_pct and policy



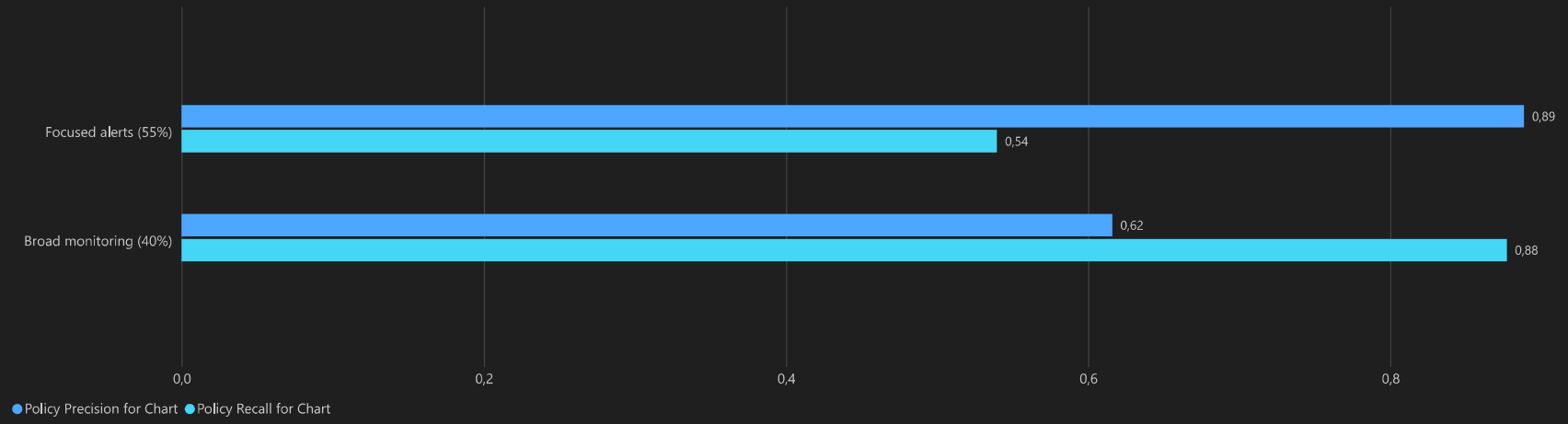
MODEL PERFORMANCE & ALERT QUALITY

Evaluating alert accuracy, risk coverage and monitoring trade-offs.

Focused Precision	Focused Recall	Broad Precision	Broad Recall
88,84 %	53,96 %	61,59 %	87,71 %

Precision vs Recall by Monitoring Policy.

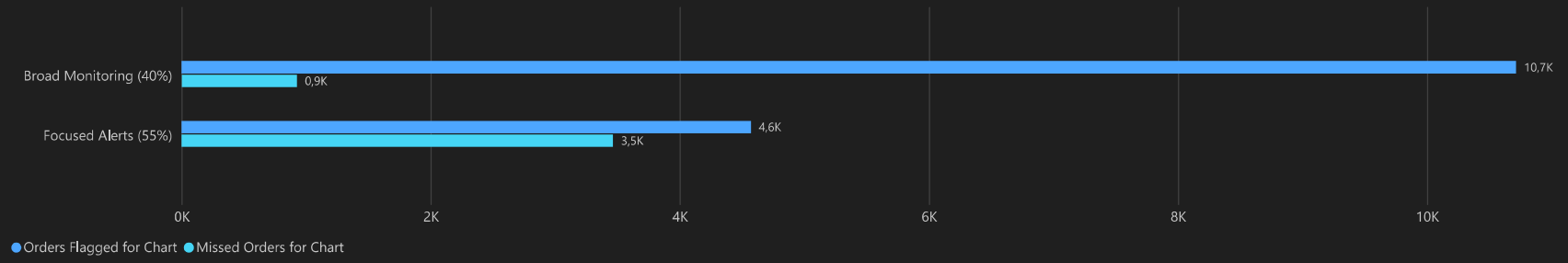
Precision, Recall, Orders flagged, and Positive-label orders missed



Focused alerts identify positive-label orders more accurately, but miss more of them. Broad monitoring captures a greater proportion of positive-label orders at the expense of more false

Alert Workload vs Missed Positive-Label Orders.

Orders Flagged for Chart and Missed Orders for Chart by Policy



Operational trade-off: Broad monitoring creates a larger alert workload but misses fewer positive-label orders. Focused alerts reduce the number of orders requiring attention, but leave more positive-label orders unflagged.